CS23431 - OPERATING SYSTEM

EXP 3: Non-Preemptive Priority Algorithm Using C

QUESTION :

Write a C program to implement the Non-Preemptive Priority Scheduling algorithm. The program should accept number of processes their Arrival Time (AT), Burst Time (BT), and Priority for each process. The process with the highest priority (lowest numerical value) must be executed first among those that have arrived. Calculate the Average Waiting Time and Average Turnaround Time.

CODE:

```
#include <stdio.h>
struct Process {
   int id, at, bt, priority;
   int ct, tat, wt, completed;
};
int main() {
   int n;
   printf("Enter number of processes: \n");
   if (scanf("%d", &n) != 1) return 0;
   struct Process p[n];
   for (int i = 0; i < n; i++) {
       p[i].id = i + 1;
       printf("Enter AT, BT and Priority for P%d: \n", p[i].id);
       scanf("%d %d %d", &p[i].at, &p[i].bt, &p[i].priority);
       p[i].completed = 0;
   }
   int current_time = 0, completed_count = 0;
   float total_tat = 0, total_wt = 0;
   while (completed_count < n) {
       int idx = -1;
       int min_priority = 1e9;
       for (int i = 0; i < n; i++) {
           if (p[i].at <= current_time && !p[i].completed) {
               if (p[i].priority < min_priority) {
                   min_priority = p[i].priority;
                   idx = i;
               } else if (p[i].priority == min_priority) {
                   if (p[i].at < p[idx].at) {
                       idx = i;
                   }
```

```
                }
            }
        }
        if (idx != -1) {
            p[idx].ct = current_time + p[idx].bt;
            p[idx].tat = p[idx].ct - p[idx].at;
}
```

Output:

Custom Test Case

Custom Test — Success

Input :

```
3
0 5 2
1 3 1
2 2 3
```

Expected Output :

```
Enter number of processes:
Enter AT, BT and Priority for P1:
Enter AT, BT and Priority for P2:
Enter AT, BT and Priority for P3:

Average Turnaround Time = 6.67
Average Waiting Time = 3.33
```

Output :

```
Enter number of processes:
Enter AT, BT and Priority for P1:
Enter AT, BT and Priority for P2:
Enter AT, BT and Priority for P3:

Average Turnaround Time = 6.67
Average Waiting Time = 3.33
```